

DOCUMENT NAME

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1. Introduction

A Test Plan is a formalized software quality document describing the scope, approach, resources, and schedule of intended testing activities. This Test Plan for the Elderly Health Monitoring and Alert System (EHMAS) outlines testing strategies for mobile and web applications developed during Capstone Project 1.

The system integrates AI fall detection, health monitoring features, alert notifications, and caregiver dashboards. Testing ensures that all functional and non-functional requirements are met according to the Product Backlog and Sprint deliverables.

1.1. Purpose:

The purpose of this Test Plan is to:

- Define all testable features of the Elderly Health Monitoring and Alert System.
- Provide a structured testing strategy covering unit, integration, system, regression, and acceptance testing.
- Identify required test resources, tools, and test environments.
- Establish a test schedule aligned with Sprint deliverables.
- Ensure product quality by validating functional requirements (User Stories) and non-functional requirements (performance, security, usability).
- Detect defects early, prevent regression issues, and ensure system readiness for final release.

This Test Plan applies to release 1.1 (as per template), which includes:

- Mobile App for Caregivers
- Web Dashboard for Admin
- AI Fall Detection Service
- Backend API & Database

1.2. References

No.	Reference	Document Information
1	Test plan Template	Mentor's template
2	Proposal	C1SE.40_Proposal_Ver_2.0.docx
3	Product Backlog	C1SE.40_ProductBacklog_SprintBacklog_Ver_2.0.xlsx
4	Project Plan	C1SE.40_Project Plan_Ver_2.0.docx
5	Test Case Document	C1SE.40_TestCase_Ver_2.0.xlsx
6	System Architecture & SRS	Sections from Proposal & Project Plan describing architecture, flows, and modules

1.3. Background information.

The Elderly Health Monitoring and Alert System (EHMAS) aims to provide:

- Continuous monitoring of elderly individuals through AI-based vision analysis and IoT sensors.
- Real-time fall detection using YOLO pose estimation + MPU6050 sensor data.
- Instant notifications to caregivers via mobile app.
- A dashboard for admins to manage users, devices, alerts, and health records.
- Health monitoring modules: BMI, Weight Tracking, Hydration Reminder, Medical Contact, Alerts, and basic AI Symptom Checker.

The project adopts the Scrum methodology with iterative development cycles. Testing activities are distributed across Sprints and executed continuously.

This Test Plan ensures that every sprint increment is thoroughly validated before integration into the final system.

1.4. Scope of testing:

Mobile App (React Native)

- User Authentication (Register, Login, Logout, Forgot Password, Verify Email)
- Home Dashboard
- Weight & BMI (PB08)
- Blood Pressure (PB11)
- Blood Sugar (PB12)
- Water Reminder (PB13)

- Alert Management (PB15)
- Medical Contact (PB16)
- Receiving Fall Alerts
- Viewing snapshots/video events

Web Dashboard (React + Tailwind)

- Admin Authentication
- User & Device Management
- Fall Event Management
- System Settings

Backend (Node.js + Express)

- API endpoints (auth, BMI, alerts, contacts, sensors, fall events)
- Authentication & Authorization (JWT)
- Database operations (MySQL)

AI Fall Detection Service (FastAPI + YOLO/OpenCV)

- Pose recognition
- Fall prediction logic
- Event trigger

IoT Sensor Integration

- ESP32 + MPU6050 sensor
- Data transmission
- Low-level fall detection

Non-Functional Tests

- Performance
- Reliability
- Security
- Usability
- Compatibility

1.5. Constraints:

Based on Project Plan + Proposal:

- Limited team experience with AI + IoT + multi-platform development.
- System performance depends on hardware (camera quality, CPU).
- Internet latency may affect real-time alerts.
- Use of free-tier cloud services may limit storage and uptime.
- Fall dataset may not fully represent real-world cases.
- Tight time constraints due to academic schedule.

1.6. Risk list:

No.	Risk	The plan for corrective	The level of influence
1	AI model inaccurate or unstable	Improve dataset, add real-world data, adjust thresholds	High
2	Internet/camera instability interrupts monitoring	Add retry logic, fallback sensor-based detection	High
3	Team members have limited experience	Mentor guidance, training sessions	Medium
4	Unexpected bugs delay sprints	Increase buffer time, daily debugging	High
5	Security vulnerabilities (JWT, API access)	Add indexing, optimize queries	Medium
6	Database overload / poor performance	Add indexing, optimize queries	Medium
7	False fall alerts annoy users	Improve logic + human confirmation	High

2. Requirements for test.

2.1. Test items:

Table 1: Test Items-App Mobile

ID_Function	Function	Type Test case	Note
APP-F01	Login	Mobile UI Test, Function Test	
APP-F02	Logout	Mobile UI Test, Function Test	
APP-F03	Register Account	Functional Test	
APP-F04	Forgot Password	Functional Test	
APP-F05	View Homepage	UI Test, Functional Test	Show BMI, Weight, Water Reminder, Alerts

APP-F06	Weight & BMI	Functional Test	Add / Edit / Delete record
APP-F07	Water Reminder	Functional Test	Setup schedule, notification
APP-F08	Fall Alert Notifications	Functional Test	Receive real-time alert
APP-F09	View Alert History	Functional Test	Include image/video snapshot
APP-F10	AI Doctor (Symptom Checker)	Functional Test	Basic Q&A
APP-F11	Medical Contact (Emergency Call)	Functional Test	Trigger call
APP-F12	Change Language	UI Test	VN / EN
APP-F13	Update Profile	Functional Test	
APP-F14	Privacy Consent	UI Test, Functional Test	Required before using app

Table 2: Test Items-Web Dashboard

ID Function	Function	Type Test case	Note
WEB-F01	Login	Web UI Test, Functional Test	Admin-only
WEB-F02	Logout	Web UI Test, Functional Test	
WEB-F03	Dashboard Overview	Functional Test	System activities, statistics
WEB-F04	Manage Users	Functional Test	CRUD, search, pagination
WEB-F05	Device Management	Functional Test	Register/approve/remove devices
WEB-F06	Registration Approval	Functional Test	Approve new users/devices
WEB-F07	Fall Alert Monitoring	Functional Test	Real-time + history
WEB-F08	Activity History	Functional Test	Filter by user / date
WEB-F09	Generate Reports	Functional Test	PDF/CSV export
WEB-F10	Settings	Functional Test	System configuration
WEB-F11	Language Setting	UI Test	VN / EN
WEB-F12	Security & Access Control	Functional Test	RBAC (admin only)

3. Test strategy

3.1. Test stages:

Test Types	Test Stages		
	Unit	Integration	Acceptance
Functional Tests	X	X	X
User Interface (UI)	x	x	x
Reliability - Integrity - Structure	X	X	
Performance Tests		X	X
Security Tests		X	X
IoT Communication Tests (ESP32 ↔ Backend ↔ App)	X	X	X
AI Model Validation (YOLO Pose + Fall Detection)	X	X	
End-to-End Workflow Tests (Camera → AI → Alert → App Notification)		X	X

3.2. Tools

Purpose	Tool	Vendor	Version
Write Test Plan, Test Cases, Test Report	Microsoft Excel	Microsoft	2021
API testing for Backend (Node.js) and AI services (FastAPI)	Postman	Postman Inc.	Latest
Automation Testing for Web Dashboard	Selenium IDE	Selenium Foundation	OpenSrc
Performance Testing (Alert response time, API load test)	JMeter	Apache	Latest
Unit Test for Node.js Backend	Jest/Mocha	Open Source	Latest
Unit Test for AI fall detection (Python)	PyTest	Open Source	Latest

4. Resource

4.1. Human Resource

The role	Resource	Specific responsibilities
Mobile App Tester	2 Tester	- Execute functional test cases for the Mobile App (Login/Signup, BMI tracking, Fall Alert, Water Reminder, AI Doctor, Vital Monitoring...) - Perform UI/UX validation for Android/iOS - Validate data synchronization between Mobile App → Backend → Admin Dashboard - Log results and defects in the defect tracking system - Re-test and verify fixed defects - Document incidents and unexpected behaviors
Web Admin Tester	1 Tester	- Execute test cases for Web Admin features (Dashboard, Device Management, User Management, Fall Event Logs, Reports...) - Verify CRUD operations for Admin modules - Validate charts, analytics, and pagination - Perform cross-browser testing (Chrome, Edge, Firefox) - Log results, defects, and generate QA reports
Integration Tester (IoT + Backend)	1 Tester / Developer support	- Test data flow between ESP32 → Backend API → System Database - Validate real-time fall detection events - Verify push notifications and alert synchronization - Test API endpoints using Postman / Swagger - Support mobile & web testers during API-related defects
Testing System / QA Environment Manager	1 Member	- Test data flow between ESP32 → Backend API → System Database - Validate real-time fall detection events

		- Verify push notifications and alert synchronization - Test API endpoints using Postman / Swagger
Testing System / QA Environment Manager	1 Member	- Set up and maintain test environments (Staging server, Test Database, API Gateway) - Support deployment for test builds (Mobile APK, Web staging) - Manage access permissions for testers - Ensure recovery and backups for test environment - Monitor test devices (phones, ESP32, camera test setup)
Automation Support (Optional)	1 Member	- Write automated UI test scripts for critical flows - Maintain Selenium / Appium automation tasks - Execute smoke tests on every new build

5. Test Milestone

Milestone Task	Name	Date Start	Date Finish
Test Plan	Entire QA Team	01/10/2025	05/10/2025
Test plan and test case for Sprint 1	Tester	06/10/2025	20/10/2025
Test plan and test case for Sprint 2	Tester	21/10/2025	10/11/2025
Test plan and test case for Sprint 3	Tester	11/11/2025	30/11/2025
Test plan and test case for Sprint 4	Tester	01/12/2025	10/12/2025

